

Graham Anderson

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I build agent systems that can prove what they did.

Principal AI Engineer | Machine Learning Engineer | AI Architect | Agentic AI · LLM/RAG · Knowledge Graphs | Defense & Aerospace | Founder, [grahamaco](#) | Buffalo, NY · Remote

U.S. citizen. Extensive experience delivering under export-controlled (ITAR) constraints.

ABOUT

An unusual résumé: commercial composer for Adidas and Pepsi, Webby-recognized producer for Sony, DARPA technical lead alongside Lockheed Martin and MIT. High-end creative and hard technical work — delivered by the same person, shipped as working code, in public.

I build verifiable agentic AI systems for defense, aerospace, and compliance. On DARPA ARCOS, I was Principal Data Scientist and ACERT Technical Lead for CS Group's contribution: the responsible technical/project lead for knowledge-graph and LLM reasoning delivery for automated certification of mission-critical software, working alongside Honeywell, Lockheed Martin, MIT, GE Research, SRI, and other program collaborators.

Much of my current client work is export-controlled, so client and program details are withheld. Publicly releasable independent engineering is at github.com/grahama1970:

- tau — receipt-gated multi-agent harness: agent work compiles to typed DAG contracts; every handoff needs a schema-valid receipt or validator result before the next step runs. No receipt, no action.
- pdf_oxide — my heavily diverged fork of yfedoseev/pdf_oxide (independent since Mar 2026): 430 commits and ~137K lines added, including Rust-core changes, a full Python pipeline/plugin system, PDF-cloning fixture generation, extraction calibration, and NIST document-validation tooling.
- agent-skills — 342 reusable agent skills (289 carrying deterministic sanity gates) and 92 bounded worker roles, portable across Claude Code, Codex, and Gemini. Public repo, private runtime — effectively my living resume.
- Private, regulated side: an ArangoDB agent-memory platform — approximately 219K evidence-grounded question–reasoning–answer records across 7K+ security-control records (NIST 800-53/171, CWE, MITRE ATT&CK, D3FEND, SPARTA), hybrid BM25 + vector + graph recall over 2.2M source chunks; a corpus of ~94K Lean 4 theorem statements indexed for requirements-to-proof retrieval.

Presented Agentic Cybersecurity research to Air Force Research Laboratory and received an AFRL "Hacker" challenge coin following the presentation.

15+ years of hand-coding underneath it all — today I work primarily through agentic coding, driving many harnesses and models, including my own (tau). I build the tools that make agentic development trustworthy, and I use them daily. Creative systems thinking — from leading 80+ person interactive productions — is why my AI systems are architected, not stapled together.

Buffalo, NY based (hybrid/onsite there, or remote). Open to contract engagements and full-time Principal/Staff AI Engineer, AI Architect, Machine Learning Engineer, LLM Platform, and Security/Compliance AI roles.

EXPERIENCE

Founder & Principal AI Engineer / Architect | [grahamaco](#) | Buffalo–Niagara Falls Area · Remote

Feb 2025 - Present Independent AI engineering practice. Clients are primarily export-controlled (ITAR) defense/aerospace — names withheld; publicly releasable engineering is at github.com/grahama1970.

- Built tau, a receipt-gated multi-agent orchestration harness: goals compile to typed DAG contracts; every agent handoff must produce a schema-valid receipt, hashed evidence artifact, or validator result — no receipt, no action.
- Develop a heavily diverged fork of pdf_oxide (origin: yfedoseev/pdf_oxide, MIT/Apache-2.0; independent since Mar 2026): 430 commits, ~137K lines added — Rust-core changes, full Python pipeline/plugin system, PDF-cloning fixture generation, extraction calibration, and NIST document-validation tooling.
- Authored agent-skills: 342 reusable agent capabilities (compliance evidence mapping, document AI, LLM evaluation, retrieval) and 92 bounded worker roles, 289 of them gated by deterministic sanity checks — public repo, private runtime.
- Built an ArangoDB agent-memory platform (private, regulated): ~219K evidence-grounded QRA records across 7K+

security-control records; hybrid BM25 + vector + multi-hop graph recall over 2.2M source chunks; ~94K Lean 4 theorem statements indexed for requirements-to-proof retrieval.

- Core contributor to pi-mono (open-source agent framework/TUI): extension system, agent runtime, and TUI harness work, Feb–Jul 2026.

Lead Research Scientist (Agentic Formal Methods) | Self-Employed / Various Clients | Buffalo, NY

Jan 2024 - Feb 2025

- Designed verifiable agentic systems: a process-driven autoformalization workflow translating engineering requirements into Lean 4 proofs for safety-critical workflows.
- Built probabilistic-deterministic self-correcting loops where LLM generation is validated by compiler/prover feedback.
- Consulted in aerospace/defense settings under export-controlled / ITAR constraints.
- Presented Agentic Cybersecurity work to Air Force Research Laboratory; received an AFRL "Hacker" challenge coin.

Principal Data Scientist & ACERT Technical Lead | CS Group (DARPA ARCOS)

Sep 2020 - Dec 2023

- Responsible technical/project lead for CS Group's contribution to the 4-year DARPA ARCOS program (Automated Rapid Certification of Software), working alongside Honeywell, Lockheed Martin, MIT, GE Research, SRI, and other program collaborators.
- Led design and delivery of ACERT (Automated Certification of Requirements Tool): architected the ArangoDB knowledge-graph schema and LLM-assisted reasoning pipeline for multi-hop compliance verification of mission-critical software against complex certification standards.
- Worked with government and aerospace program stakeholders; split time roughly 50/50 between executive/technical briefings and hands-on architecture and code.

Executive Producer / Marketing Executive / Professional Composer | Los Angeles, CA

Mar 2005 - 2016

- Executive Producer, God of War: Ascension interactive campaign (Real Pie, for Sony; Webby-recognized); led cross-functional teams (80+) on large-scale interactive productions.
- Commercial composer for major consumer brands (Adidas, X-Games, Pepsi, Taco Bell, Nestle).

PUBLIC WORK (non-ITAR) — github.com/grahama1970

Client work is mostly export-controlled, so here is the public, verifiable side — including the fun stuff:

- **agent-skills** — my living resume: 342 reusable agent skills, 92 worker roles, 289 with sanity gates. Public repo, private runtime.
- **tau** — receipt-gated multi-agent harness. "Agents hallucinate. Tau contains them."
- **pdf_oxide** — heavily diverged fork of yfedoseev's Rust PDF toolkit (430 commits, ~137K lines added: Rust-core changes, Python pipeline, PDF cloning, NIST validation).
- **scillm**, **extractor**, **anvil**, **fetcher**, **chatterbox** voice-agent fork — supporting cast, all public.

TOP SKILLS

Machine Learning, Generative AI, Large Language Models (LLM), Agentic AI, AI Agents, Multi-Agent Systems, Retrieval-Augmented Generation (RAG), Knowledge Graphs, GraphRAG, ArangoDB, Vector Databases, Prompt Engineering, LLM Evaluation, LLM Ops, Formal Verification, Lean 4, Python, Rust, Model Context Protocol (MCP), NIST Frameworks, AI Governance, Docker, Linux